

Performance Testing Report

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Project Title

Performance Testing of Dashboard Feature using Apache JMeter

Introduction

Performance testing is a type of software testing used to measure the speed, stability, responsiveness, and reliability of an application under different workloads. It helps identify performance bottlenecks and ensures that the system performs efficiently when multiple users access it simultaneously.

For this task, Apache JMeter was used to perform load testing on the dashboard feature of the sample application hosted on httpbin.org.

Objective

The objective of this performance testing activity was:

- To understand the basics of performance and load testing
- To simulate multiple virtual users
- To measure response time and throughput
- To analyze system behavior under load
- To generate reports and graphs using Apache JMeter

Tool Used

Tool	Version
Apache JMeter	5.6.3

Test Environment

Component	Details
Operating System	Windows 10
Testing Tool	Apache JMeter
Website Tested	httpbin.org
Protocol Used	HTTPS
Browser	Google Chrome
Internet Connection	Wi-Fi

Test Scenario

A load test was designed in Apache JMeter to test the dashboard feature of the sample application.

The following JMeter components were used:

- Thread Group
- HTTP Request Sampler
- Summary Report
- Graph Results
- View Results Tree
- Response Time Graph

The test simulated multiple users sending requests to the server simultaneously.

Load Configuration

Parameter	Value
Number of Virtual Users	50
Ramp-Up Period	5 Seconds
Loop Count	1
Request Method	GET

Performance Metrics

The following performance metrics were analyzed during testing:

1. Response Time

Measures how long the server takes to respond to requests.

2. Throughput

Measures the number of requests processed per second.

3. Error Rate

Shows the percentage of failed requests during testing.

4. Latency

Measures the delay between request submission and response start.

Test Execution

The test plan was executed successfully in Apache JMeter. Multiple virtual users accessed the dashboard feature simultaneously. Reports and graphs were generated to analyze application performance.

Observations

- The application responded successfully to requests.
- Response times remained stable during execution.
- No major performance issues were observed.
- The server handled multiple requests efficiently.
- Error rate was very low or zero.

Results Summary

Metric	Observation
Average Response Time	Stable
Error Rate	0%
Throughput	Good
System Stability	Acceptable

Graph Analysis

The generated graphs and reports showed:

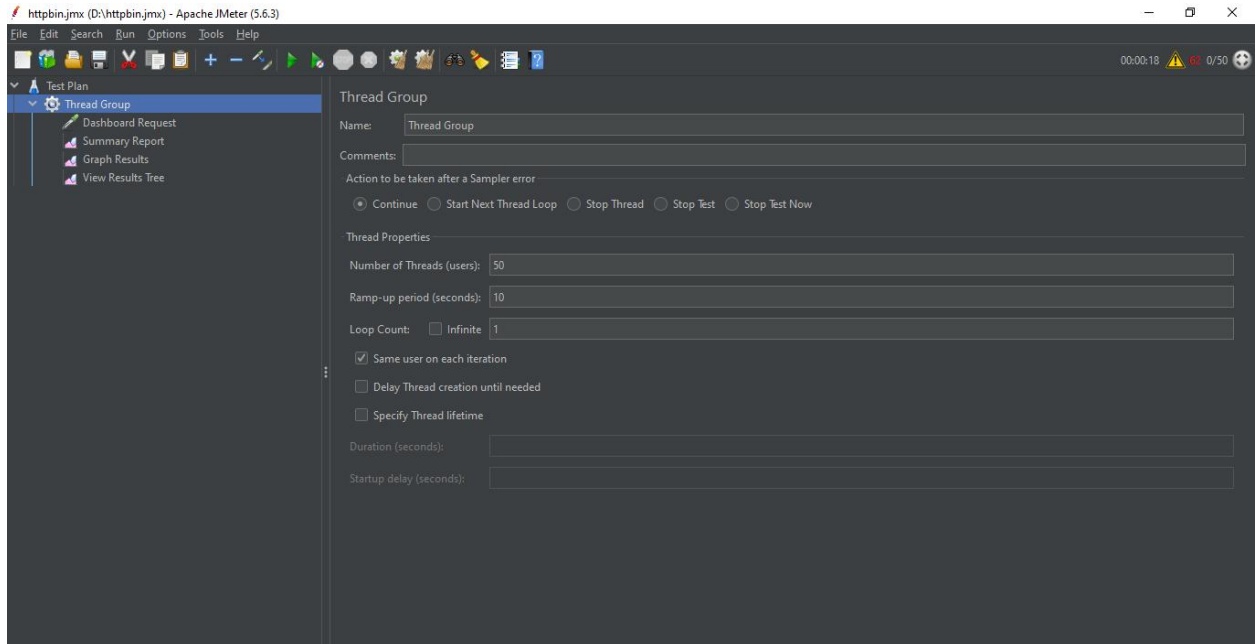
- Stable response times during testing
- Successful request execution
- Consistent server performance
- Smooth handling of multiple users

The Summary Report and Graph Results confirmed that the application performed efficiently under the applied load.

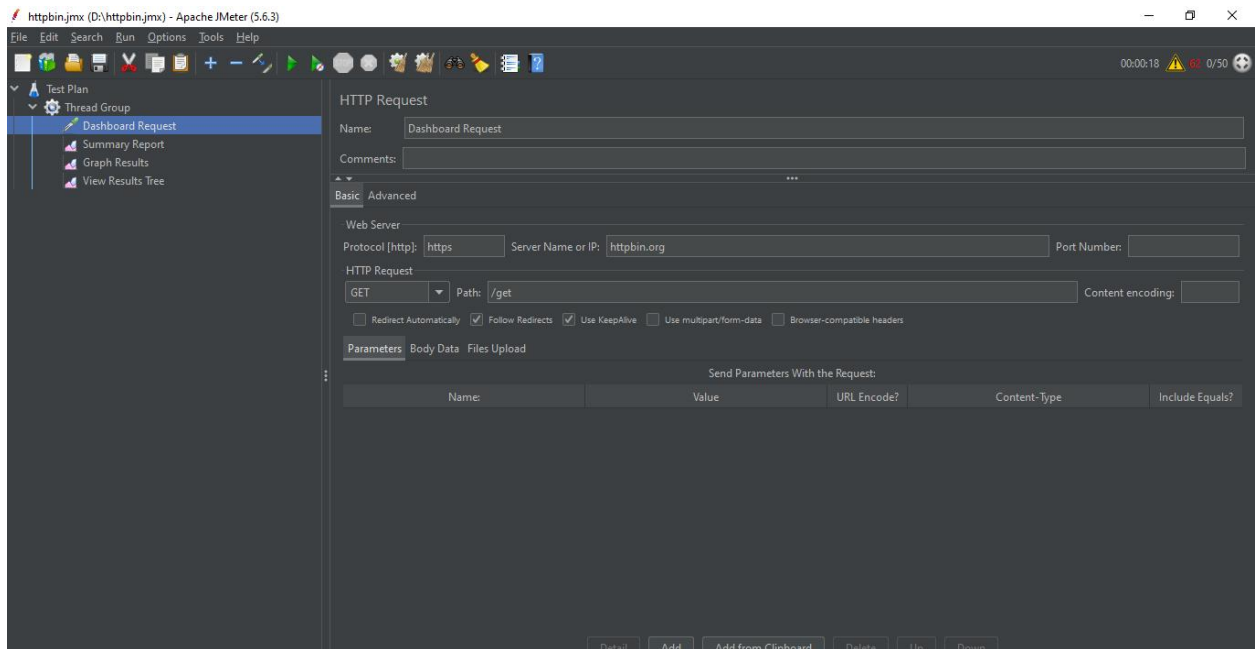
Screenshots Included

The following screenshots were attached in the report:

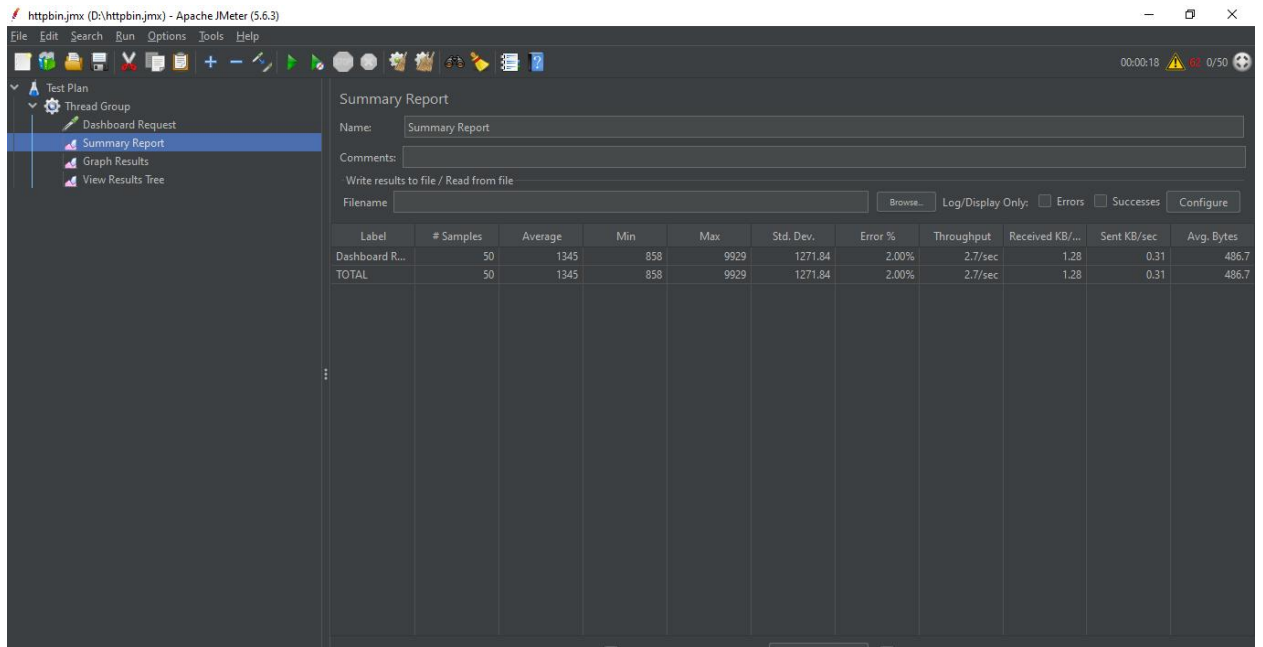
1. Thread Group Configuration



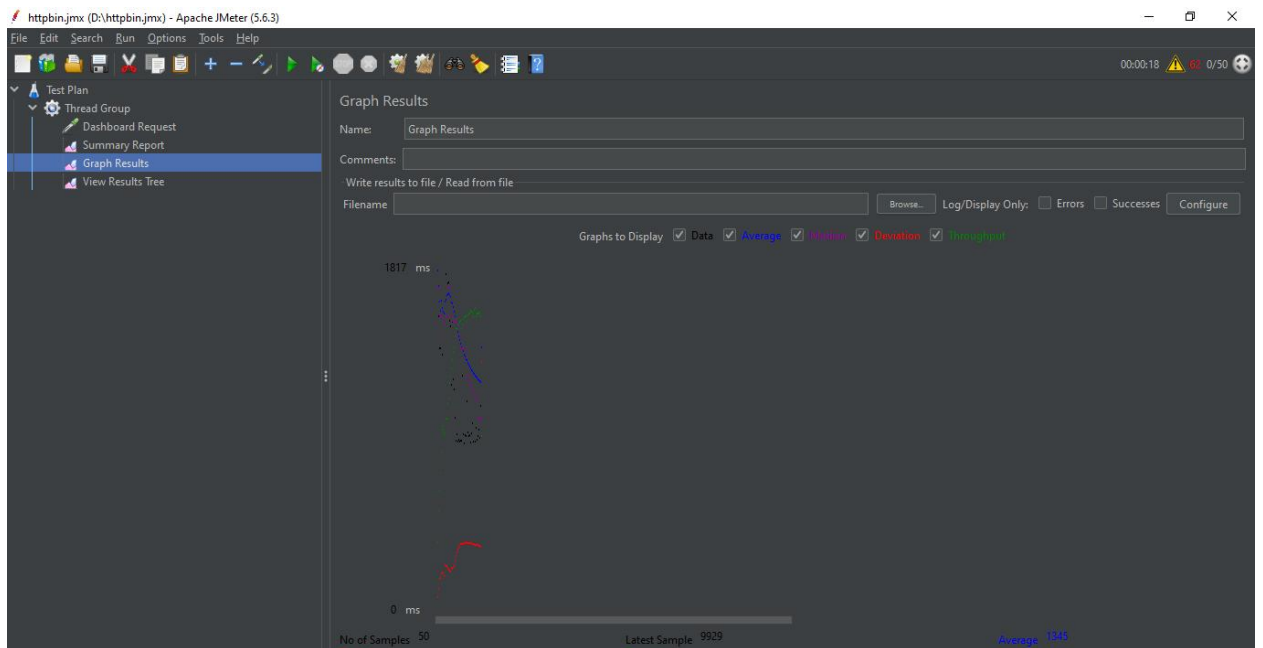
2. HTTP Request Setup (Dashboard request)



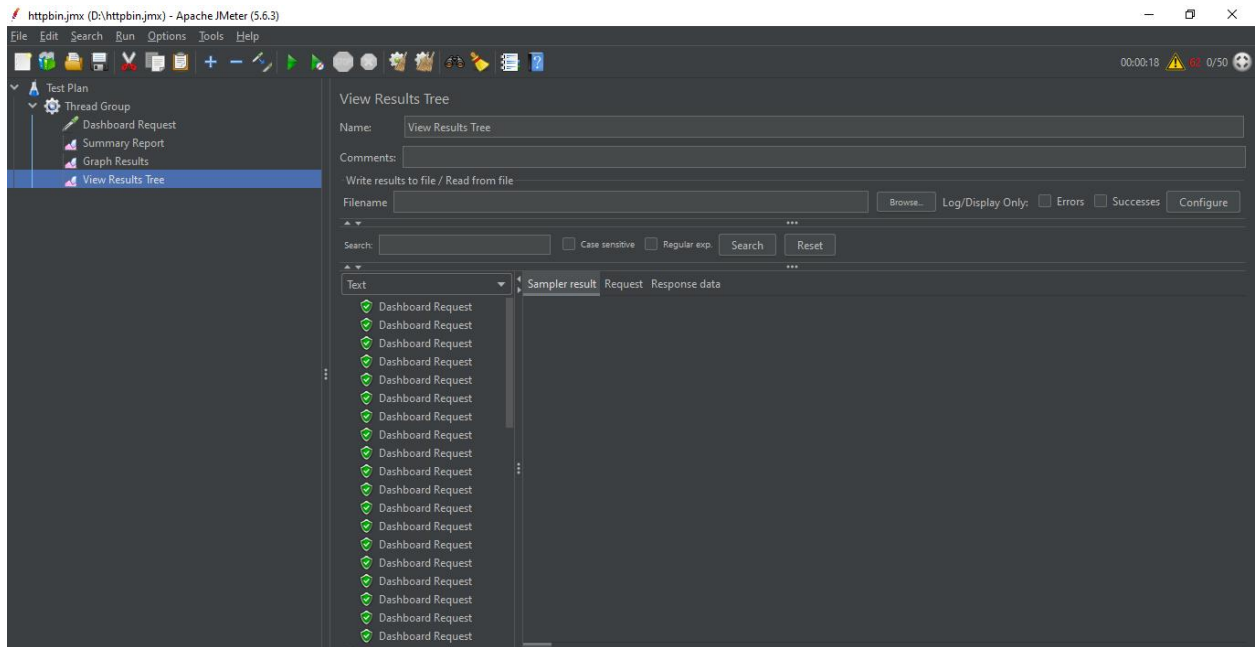
3. Summary Report



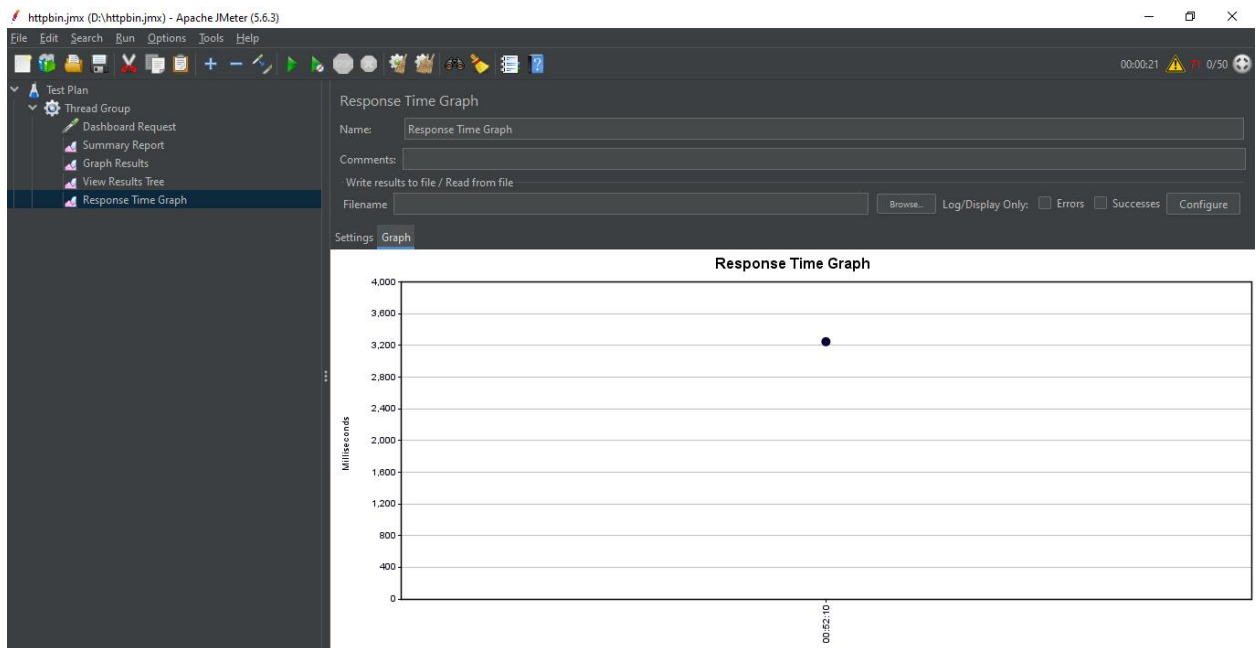
4. Graph Results



5. View Results Tree



6. Response Time Graph



Conclusion

The dashboard feature of the sample application was tested successfully using Apache JMeter. The application handled multiple virtual users efficiently with stable response times and minimal errors. The performance of the application under load was acceptable.

Recommendations

- Increase the number of virtual users for stress testing.
- Perform testing in a production-like environment.
- Monitor CPU and memory usage during testing.
- Use advanced test scenarios with larger datasets.
- Conduct regular performance testing for system optimization.